

COMBINATORICS EXAM 2

ROB TOMPKINS

1. (25pts): Let a_n be the number of ways to partition an n -element set into ordered pairs (i.e. matchings where order within each pair matters). Find the exponential generating function for a_n . Derive a closed-form expression for a_n .

Solution. The pairs themselves are unordered, but the two entries in each pair are ordered. On a fixed two-element set there are $2!$ ordered pairs, so the exponential generating function for one pair is $2!x^2/2! = x^2$. The exponential formula therefore gives

$$A(x) = \sum_{n \geq 0} a_n \frac{x^n}{n!} = \exp(x^2) = \sum_{m \geq 0} \frac{x^{2m}}{m!}.$$

Comparing coefficients yields

$$a_{2m} = \frac{(2m)!}{m!}, \quad a_{2m+1} = 0.$$

Here $a_0 = 1$, corresponding to the empty partition. Alternatively, arrange the $2m$ elements in a list and split it into consecutive pairs. Each partition is counted $m!$ times, once for each ordering of its pairs, giving the same formula. $\square$

2. (25pts): Let a_n be the number of ways to distribute n labeled objects into any number of labeled boxes such that each box contains at least two objects. Construct the exponential generating function for a_n . Extract a formula for a_n .

Solution. Interpret “any number of labeled boxes” as summing over $k \geq 0$, with the boxes labeled $1, \dots, k$ for each fixed k . (Choosing arbitrary labels from an infinite supply would not give a finite count.) A single box contains an unordered set of at least two labeled objects, with exponential generating function

$$B(x) = \sum_{j \geq 2} \frac{x^j}{j!} = e^x - 1 - x.$$

For k labeled boxes the generating function is $B(x)^k$. Consequently, as a formal power series,

$$A(x) = \sum_{k \geq 0} (e^x - 1 - x)^k = \frac{1}{2 + x - e^x}.$$

If box i contains n_i objects, the number of distributions is the multinomial coefficient $n!/(n_1! \cdots n_k!)$. Thus an explicit finite formula is

$$a_n = \sum_{k=0}^{\lfloor n/2 \rfloor} \sum_{\substack{n_1 + \cdots + n_k = n \\ n_i \geq 2}} \frac{n!}{n_1! \cdots n_k!}.$$

The $k = 0$ term is 1 when $n = 0$ and 0 otherwise. Equivalently, $a_n = \sum_k k! \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}_{\geq 2}$, where the symbol counts partitions into k blocks, each of size at least two. $\square$

3. (25pts): The Catalan numbers are defined by $C_n = \frac{1}{1+n} \binom{2n}{n}$. Probe that C_n satisfies the recurrence $C_n = \sum_{k=0}^{n-1} C_k C_{n-1-k}$. Give a combinatorial interpretation of this recurrence.

Proof. Let $C(x) = \sum_{n \geq 0} C_n x^n$. The formal binomial expansion gives

$$\frac{1 - \sqrt{1 - 4x}}{2x} = \sum_{n \geq 0} \left[-\frac{1}{2} \binom{1/2}{n+1} (-4)^{n+1} \right] x^n = \sum_{n \geq 0} \frac{1}{n+1} \binom{2n}{n} x^n.$$

Hence $C(x) = (1 - \sqrt{1 - 4x})/(2x)$, which satisfies $C(x) = 1 + xC(x)^2$. Taking the coefficient of x^n , for $n \geq 1$, proves

$$C_n = \sum_{k=0}^{n-1} C_k C_{n-1-k}, \quad C_0 = 1.$$

Combinatorially, C_n counts Dyck paths: paths with n up-steps and n down-steps that never go below height zero. Indeed, the reflection principle counts the paths that go below zero as $\binom{2n}{n+1}$, so the number of Dyck paths is $\binom{2n}{n} - \binom{2n}{n+1} = C_n$. Every nonempty Dyck path decomposes uniquely as $UPDQ$, where the displayed D is the first return to height zero. If P has semilength k , then Q has semilength $n - 1 - k$. There are $C_k C_{n-1-k}$ choices for these two paths; summing over k gives the recurrence. $\square$

4. (25 pts): How many integers between 1 and 1000 (inclusive) are divisible by at least one of 3, 5, or 7? Use the Inclusion-Exclusion Principle and show all steps.

Solution. Let $A_d = \{m \in \{1, \dots, 1000\} : d \mid m\}$. Since 3, 5, 7 are pairwise coprime, their intersections consist of multiples of the corresponding products. Thus

$$\begin{aligned} |A_3| &= 333, & |A_5| &= 200, & |A_7| &= 142, \\ |A_3 \cap A_5| &= \lfloor 1000/15 \rfloor = 66, \\ |A_3 \cap A_7| &= \lfloor 1000/21 \rfloor = 47, \\ |A_5 \cap A_7| &= \lfloor 1000/35 \rfloor = 28, \\ |A_3 \cap A_5 \cap A_7| &= \lfloor 1000/105 \rfloor = 9. \end{aligned}$$

By inclusion–exclusion,

$$\begin{aligned} |A_3 \cup A_5 \cup A_7| &= 333 + 200 + 142 - 66 - 47 - 28 + 9 \\ &= \boxed{543}. \end{aligned}$$

□

5. (25pts): Suppose we have an infinite supply of pennies and we would like to stack them up in rows. Find the number of different arrangements possible subject to the following conditions:

- (i) the bottom row consists of n pennies, each touching its two neighbors, except the ends where $n \geq 1$;
- (ii) a coin that does not belong to a row sits on two coins below it;
- (iii) do not distinguish between heads and tails.

Solution. Interpret condition (ii) as saying that every coin outside the *bottom* row rests on two adjacent coins immediately below it. Upper rows may have gaps, and left–right reflections are counted separately. Let f_n count these stacks on a fixed bottom row of n coins, and set $f_0 = 1$.

There are $n - 1$ possible coin positions immediately above the bottom row. If this entire next row is occupied, the choices above it form a stack with bottom row of length $n - 1$, giving f_{n-1} arrangements.

Otherwise, let k be the first vacant position in that row, counted from the left, where $1 \leq k \leq n - 1$. The first $k - 1$ positions are occupied, and the choices above these coins give f_{k-1} stacks. To the right of the vacancy there is an arbitrary stack on the last $n - k$ bottom coins, giving f_{n-k} choices. No coin can bridge the vacancy: any such coin would require, through its supports, the missing coin. The two choices are therefore independent. Consequently,

$$f_n = f_{n-1} + \sum_{k=1}^{n-1} f_{k-1} f_{n-k} = \sum_{k=1}^n f_{k-1} f_{n-k}.$$

This is the Catalan recurrence with $f_0 = 1$, so

$$f_n = C_n = \frac{1}{n+1} \binom{2n}{n}.$$

For example, $f_1 = 1$, $f_2 = 2$, and $f_3 = 5$. □

6. (25pts): Consider the colorings of the vertices of a regular hexagon using 3 colors. Determine the cycle index polynomial of the dihedral group D_6 . Use Polya’s Enumeration Theorem to compute the number of distinct colorings.

Solution. □

7. (25pts): Model a benzene ring as a hexagon. Suppose we substitute hydrogen atoms with two types of atoms (A and B). Using Polya's Enumeration Theorem, determine the number of distinct substitution patterns (up to rotational and reflectional symmetry).

Solution.

□

8. (25pts): Show that: $R(3, 5) > 13$ and $R(3, 5) \leq 14$.

Proof.

□